The ϕ^4 field, and the answers to 25 September

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Call with Jérôme, 19 October 2026

In one page

What we did. We built the second laboratory of your email: exact snapshots of the one-dimensional ϕ^4 (stochastic Allen–Cahn) field, six settings across temperature and lattice spacing, and the four arms you proposed (EM with a learned pair energy, the Gaussian closure, a window CNN, the exact score), plus a transformer, a neural (MDN) kernel and the paper's own mixture kernel. EM–BP learns from noisy snapshots only; the networks are trained both on noisy snapshots and, with more information, on clean ones. Sixteen seeds; $M = 32$ to 2048 sequences (networks at 128 and 2048). Each experiment was registered with its predictions before the main run, and its outputs are frozen by checksum. Predictions that failed are reported as such (*in italics*).

What came out.

1. **The five-parameter site energy wins everywhere.** Your pair energy, a Gaussian bond plus a quartic potential per site, learned by EM–BP through the noise, is the best arm on every cell at every M (on average over seeds). At $M = 2048$ its error is 0.032–0.169 of the Gaussian closure's, and the best network is 14–24 times worse.
2. **Its parameters are the physics.** The fitted temperature is within 10% of the truth in 96 of 96 fits, the barrier height within 20% in 78 of 80, and the learned potential is the double well (Figure 3). Samples drawn from it have the right kinks and domain lengths.
3. **The kernel family is the modelling choice.** The paper's mixture kernel, whose expected next value is linear in the current one, learns nothing beyond the Gaussian closure (1.004–1.021 times its error). A neural kernel earns its parameters only where the simple family is wrong (Section 5).
4. **Your refinement prediction holds.** As h goes from 1 to $\frac{1}{4}$ at fixed physical length, the window CNN goes from 0.55 to 1.20 times the Gaussian closure's error (its radius re-chosen on each lattice), while the site energy stays at 3–4% of it.

What is weaker than it sounds (Section 6). BP is slower per sequence than the networks; the stopping point of every arm, networks included, is chosen with the exact score on a validation set (a likelihood-based rule gives 4–23% instead of 3–17%); with $M = 32$ on the two hottest cells the site energy is still above the Gaussian closure; several neural-kernel numbers are upper bounds.

Table 1: Your email, point by point.

Your point	What we did, and what came out
Snapshots are exactly Markov; grid BP is an exact reference	Yes. The oracle on the 401-point grid agrees with a four times finer grid to 1.3×10^{-12} on every cell.
The mixture-of-noise family is misspecified; a learned kernel is needed	Confirmed: the mixture sits on the Gaussian closure, 1.004–1.021 times it.
Parametrise a pair energy W_θ; Boltzmann learning; the E-step is the Fisher identity	Done, with five parameters (Section 2); best arm everywhere. One addition: the directed kernel and the symmetric pair energy are the same model class (every such kernel is reversible).
Gaussian closure should fail badly	It fails, and more than expected: moment-projected BP is not even well-posed here (Section 2). The Gaussian answer is the linear floor: 0.27 on the base cell against 0.03 for the site energy.
Choose h near the correlation length, or treat h as a parameter	Two axes: a temperature sweep at fixed correlation length ($\xi \approx 6$ sites), and a refinement $h = 1, \frac{1}{2}, \frac{1}{4}$ at fixed physical length; plus a fixed-h sweep for the physics.
Fixed-window networks lose under refinement; BP does not care	Confirmed (item 4 above, Figure 4).
Domain length grows like $e^{\Delta E/\Theta}$; no transition in 1D	Smooth crossover, ξ from 1.06 to 19.5 sites as Θ goes from 2 to 0.2. *The fitted kink energy is* 0.988*, not* $2\sqrt{2}/3 \approx 0.943$: these temperatures are not cold enough.
Use the temperature sweep to study speciation	Done (Figure 5): colder fields commit earlier, and earlier than their Gaussian twin.
Exact sampling, no equilibration issues	Data are exact steps of the true kernel (rejection from the tilted walk), the first site after a burn-in that forgets the start to 10^{-12}; an inverse-CDF sampler on the transfer matrix agrees.
Optimise grid BP; Fourier not so promising	Agreed. The cheap lever is a coarser grid: on 101 points the ϕ^4 kernels lose nothing and BP is 6.2 times faster than on 401.
Frame the theory neutrally, with the linear chain as a first example	Drafted this way (general Markov kernel; linear chain first, field second); to be cut down.

1 The field as a Markov chain

The SPDE and its stationary law (we write Θ for the temperature, because T is the transfer operator):

$$\partial_\tau \phi = \partial_x^2 \phi - V'(\phi) + \sqrt{2\Theta}\,\eta, \qquad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2, \qquad p[\phi] \propto \exp\Big(- \tfrac{1}{\Theta} \int \big[\tfrac{1}{2}(\partial_x \phi)^2 + V(\phi) \big]\, \mathrm{d}x \Big).$$

On a lattice of spacing h the energy couples only neighbours,

$$p(\phi_1, \ldots, \phi_L) \;\propto\; \prod_i \exp\Big(- \frac{(\phi_{i+1} - \phi_i)^2}{2\Theta h} - \frac{h\, V(\phi_i)}{\Theta} \Big),$$

so a snapshot is exactly a Markov chain in the site index. With the symmetric transfer operator $\mathsf{T}(\phi, \phi') = e^{-hV(\phi)/2\Theta}\, e^{-(\phi'-\phi)^2/2\Theta h}\, e^{-hV(\phi')/2\Theta}$, its Perron pair $\mathsf{T}\psi_0 = \lambda_0 \psi_0$, the stationary chain has

$$K(\phi' \mid \phi) = \frac{\mathsf{T}(\phi, \phi')\, \psi_0(\phi')}{\lambda_0\, \psi_0(\phi)} = \frac{\mathcal{N}(\phi'; \phi, \Theta h)\; g(\phi')}{Z(\phi)}, \qquad g = \psi_0\, e^{-hV/2\Theta}, \qquad \pi = \psi_0^2.$$

The second form is your ground-state transform written as a *tilted walk*: a Gaussian step re-weighted by a function of the arrival point, with Z its normaliser. Two facts follow and matter below:

- **The drift is non-linear.** In standardised units $a = \phi/s$ the step variance is $\gamma = \Theta h / s^2$, and differentiating Z gives $\mathbb{E}[a' - a \mid a] = \gamma\, (\log Z)'(a)$. The paper's mixture kernel $a' = \rho a + \varepsilon$, $\varepsilon \sim \sum_c w_c \mathcal{N}(\mu_c, \sigma_c^2)$, has $\mathbb{E}[a' \mid a] = \rho a$+const, so it cannot hold a site inside a domain. The two families meet only at the Gaussian linear chain.

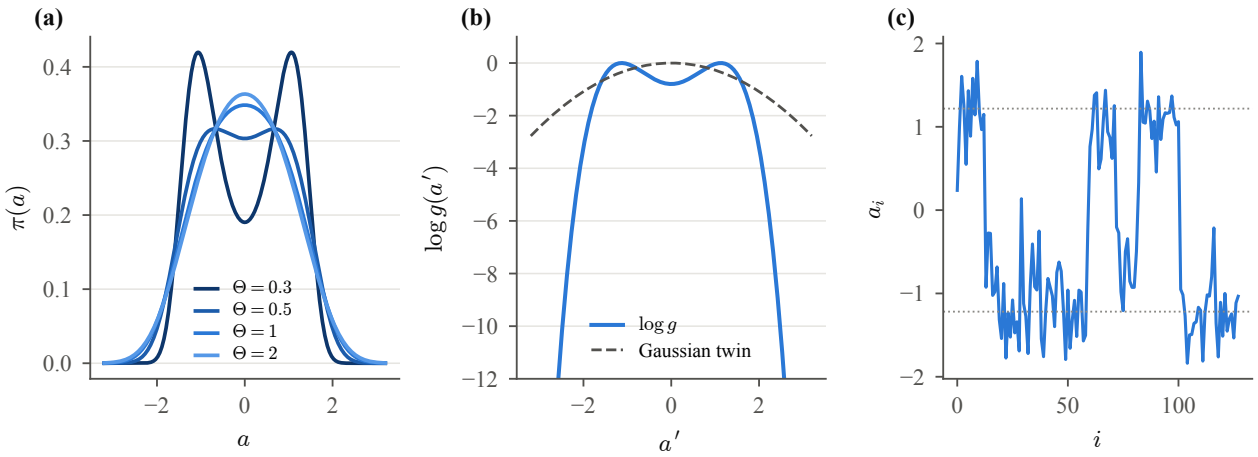


Figure 1: (a) Stationary density of the standardised field $a = \phi/s$ at the four temperatures of the sweep: two humps when cold. (b) The log-tilt $\log g$ against the Gaussian tilt of the covariance twin: bimodal, which no linear-Gaussian kernel can express. (c) An exact snapshot of 128 sites at $\Theta = 0.3$, $h = 1$; dotted lines mark the wells.

Table 2: The six cells of the field laboratory: temperature Θ, lattice spacing h (solved on the three hotter cells to hold the correlation length fixed), chain length L, correlation length ξ in sites, excess kurtosis of the marginal (kurt.), and the schedule-pooled relative score error of the best linear estimator (floor) and of the isotropic score (iso.), mean of two replicates.

Θ	h	L	ξ	kurt.	floor	iso.
0.3	1.000	32	6.1	−1.35	0.269	0.674
0.5	0.419	32	6.2	−0.88	0.115	0.657
1.0	0.226	32	6.2	−0.59	0.058	0.657
2.0	0.150	32	6.2	−0.46	0.041	0.656
0.3	0.500	64	11.5	−1.25	0.162	0.714
0.3	0.250	128	22.5	−1.22	0.098	0.750

- **Correlation length** $\xi = 1/\log(\lambda_0/\lambda_1)$ in sites; it grows like $1/h$ under refinement and roughly like $e^{E_{\text{kink}}/\Theta}$ as the field cools. There is no transition at any $\Theta > 0$.

Cells. The base cell ($\Theta = 0.3$, $h = 1$, $L = 32$) is cold enough for domains. Three hotter cells keep the correlation length at about six sites by solving for h, so they match the linear-chain laboratory at second order. Two refinements of the base cell keep the physical length fixed. The "floor" column is the error of the best linear estimator, i.e. how much there is to gain beyond a Gaussian model.

2 The estimator: your pair energy, learned through the noise

Score from BP. Data are corrupted as $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}$, $\Delta_t = 1 - e^{-2t}$. By Tweedie, $\nabla_{\mathbf{x}} \log P_t(\mathbf{x}) = -(\mathbf{x} - e^{-t}\mathbb{E}[\mathbf{a} \mid \mathbf{x}])/\Delta_t$, and on a chain the posterior mean is exact by one forward and one backward BP sweep through K, with messages on a 401-point grid on $[-8, 8]$.

The model. On the standardised field the pair energy is a Gaussian bond plus a site energy,

$$W_\theta(a, a') = \frac{(a' - a)^2}{2\gamma} + f_\theta(a) + f_\theta(a'), \qquad f_\theta(a) = \sum_{m=1}^{4} c_m \left(\frac{a}{2}\right)^m, \qquad \theta = (\log\gamma, c_1, \ldots, c_4),$$

five parameters. We call this arm the *site energy*, after its free function f_θ; it is the pair energy of your email. The kernel is the ground-state transform of $\mathsf{T}_\theta(a, a') = e^{-f_\theta(a)}\,\mathcal{N}(a'; a, \gamma)\,e^{-f_\theta(a')}$ (one eigensolve per evaluation). The truth is in the family: $\gamma = \Theta h/s^2$ and $f = (h/2\Theta)\,V(sa)$+const.

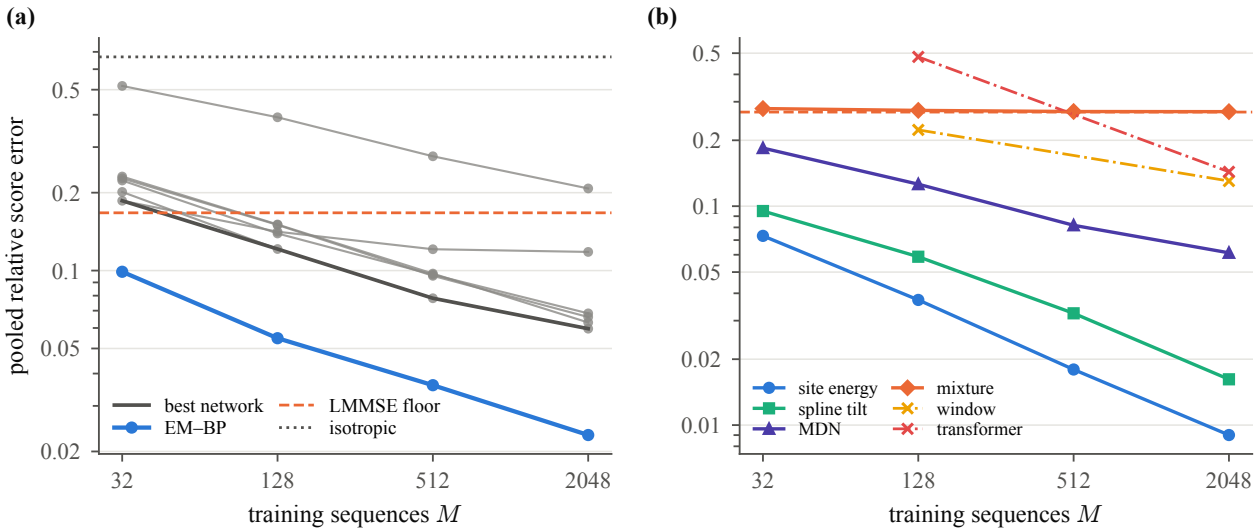


Figure 2: Pooled error against the number of training sequences. (a) The paper's linear chain with Laplace driving noise: EM–BP with the 25-number mixture kernel against the best network (trained on clean sequences; the other architectures in grey), the LMMSE floor and the isotropic score. (b) The ϕ^4 base cell ($\Theta = 0.3$, $h = 1$): the site energy improves steadily, the paper's mixture stays on the dashed linear floor, the networks (DSM) start above it.

Learning = Boltzmann learning with a posterior. For one noisy sequence, Fisher's identity gives

$$\nabla_\theta \log p_\theta(\mathbf{x}) = \mathbb{E}_{p_\theta(\mathbf{a}|\mathbf{x})}\Big[\nabla_\theta \log p_\theta(\mathbf{a})\Big] = -\,\mathbb{E}_{\text{posterior}}\Big[\textstyle\sum_i \nabla_\theta W_\theta(a_i, a_{i+1})\Big] + \mathbb{E}_{\text{model}}\Big[\textstyle\sum_i \nabla_\theta W_\theta(a_i, a_{i+1})\Big],$$

for the Gibbs form $p_\theta(\mathbf{a}) \propto \exp(-\sum_i W_\theta(a_i, a_{i+1}))$: your $\mathbb{E}_{\text{data}}[\nabla W] - \mathbb{E}_{\text{model}}[\nabla W]$ with the data term replaced by a posterior expectation. EM does this in two steps: the E-step runs BP on each noisy sequence and accumulates the expected pair counts $\boldsymbol{\Xi}(a, a')$; the M-step increases $Q(\theta) = \langle \boldsymbol{\Xi}, \log K_\theta \rangle$ (L-BFGS, gradient through the perturbation theory of the Perron vector; not concave, so this is generalised EM). EM's stopping point is chosen on a validation set (Section 6).

Directed or symmetric? The same class. Any tilted walk $K_g(a' \mid a) = \mathcal{N}(a'; a, \gamma)\; g(a')/Z_g(a)$ is reversible, with invariant law $\pi_g \propto g\, Z_g$, because $\pi_g(a) K_g(a' \mid a) \propto g(a)\, g(a')\, \mathcal{N}(a'; a, \gamma)$ is symmetric. It is the Gibbs chain of the pair energy above with $f = -\frac{1}{2}\log(g/Z_g)$. So the choice is only about parametrisation: a flexible tilt $\log g$ (we run a spline, 25 numbers, and a Legendre polynomial, 7) or the site energy itself (5, a quartic for the field). A fitted kernel then reads as physics: $\hat{\Theta} = \hat{\gamma} s^2/h$ and $\hat{V}(\phi) = (2\hat{\Theta}/h)\hat{f}(\phi/s)$+const.

What the Gaussian closure is here. Forcing the messages to be Gaussian (moment projection, which on a linear chain is exactly the Kalman smoother) is not a usable estimator on the field: far from the bulk the kernel jumps back towards it almost regardless of where it starts, so the backward messages are nearly flat across the grid and their projection discards them (0.42 of a projected message's mass in the outer 5% of the grid on the base cell, 0.98 or more on the others). The Gaussian arm is therefore the covariance twin's exact answer, the best linear estimator, computed in closed form: the "linear floor".

Arms. EM–BP with five kernels (site energy 5, Legendre tilt 7, spline tilt 25, paper's mixture 25, MDN 1,516); window CNN and transformer, each trained two ways, on one noisy copy per sequence (ISM, EM–BP's own information) and on clean targets with one fixed corruption (DSM, strictly more information); the linear floor; the exact oracle. The error is $R = \big(\sum \|\hat{S} - S\|^2 / \sum \|S\|^2\big)^{1/2}$, pooled over test chains, sites and twelve levels $t \in [0.05, 3]$; tables report R divided by the linear floor, so 1 is the Gaussian closure.

3 Results

Reading Table 3.

- **The site energy is best on every cell.** At $M = 2048$: 0.032–0.169 of the linear floor, and 0.41–0.57 of

Table 3: Risk divided by the linear floor (a Gaussian closure scores 1), mean over sixteen seeds, of every estimator on the six cells; the first four columns are the temperature sweep at matched correlation length, the last two refine the base cell at fixed physical length. EM's stopping point and each network's checkpoint are selected by the held-out risk against the exact score. A superscript counts failed fits, excluded from the mean; a dash means all sixteen failed. Bold: smallest in its column.

M	estimator	$\Theta = 0.3$	0.5	1	2	$h = 0.5$	$h = 0.25$
128	site energy (5)	**0.139**	**0.269**	**0.534**	**0.630**	**0.136**	**0.130**
	spline tilt (25)	0.218	0.454	1.111	1.478	0.231	0.235
	Legendre tilt (7)	0.394	0.354	0.647	0.816	0.330	0.387^{6}
	MDN	0.468	0.900	1.724	2.457	0.476	0.585
	mixture (paper)	1.018	1.039	1.187	1.279	1.016	1.017
	window, ISM	1.026	4.396	6.264	8.735	1.495	2.237
	transformer, ISM	3.284	9.752	8.509	14.487	3.591	4.729
	window, DSM	0.829	2.339	4.149	5.786	1.125	1.580
	transformer, DSM	1.787	4.359	8.034	11.571	1.887	2.129
2048	site energy (5)	**0.033**	**0.072**	**0.141**	**0.169**	**0.040**	**0.032**
	spline tilt (25)	0.060	0.134	0.280	0.429	0.073	0.068
	Legendre tilt (7)	0.074	0.092	0.165	0.243	0.111	–
	MDN	0.228	0.422	0.737	0.837	0.310	0.420
	mixture (paper)	1.004	1.004	1.014	1.021	1.010	1.016
	window, ISM	0.552	1.284	2.457	3.559	0.776	1.196
	transformer, ISM	1.354	2.706	2.440	3.237	1.368	1.020
	window, DSM	0.484	1.174	2.287	3.334	0.701	1.030
	transformer, DSM	0.533	1.225	2.156	3.043	0.622	0.780

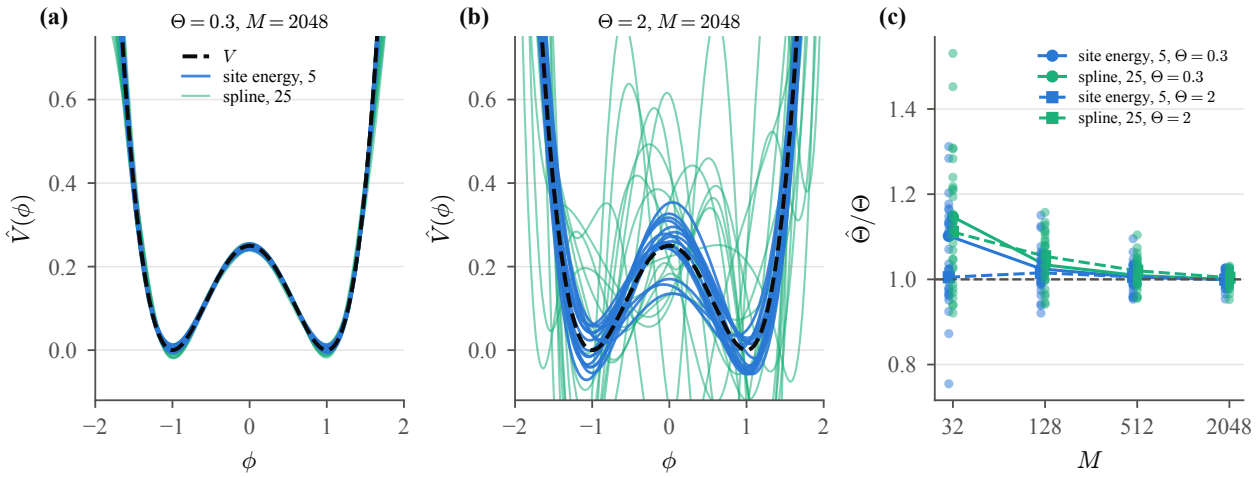


Figure 3: The potential read off the kernels learned through noise, $\hat{V}(\phi) = (2\hat{\Theta}/h)\hat{f}(\phi/s)$, one curve per seed (sixteen), against the true double well (dashed), at $M = 2048$. (a) Base cell: both kernels recover the well. (b) $\Theta = 2$, where the barrier is small against the thermal energy: the site energy keeps the right shape with a spread barrier; the 25-number spline tilt does not pin the potential down. (c) Fitted temperature over true against M, every seed as a point, on the two cells.

the 25-number spline's error. *We had predicted it would be within* 10% *of the spline; it is about twice as good.* The right five parameters beat a flexible 25.

- **The paper's mixture learns nothing beyond the floor**, at every M (Figure 2b): a linear drift cannot hold a domain. Its samples show it, 1.41–1.42 times too many kinks on the cold cells and domains 14%–18% too short at the median, while the site energy's samples are within 2.1% in kink density and 5.6% at every domain-length quantile, indistinguishable from exact samples.
- **The networks lose to the structured kernels everywhere**: at least 4.1 times the spline's error, and above the linear floor on every cell with $\Theta \geq 0.5$, whichever training recipe. They do beat the mixture on the cold cells. The MDN kernel is below the floor at $M = 2048$ on every cell, but well above the tilts.

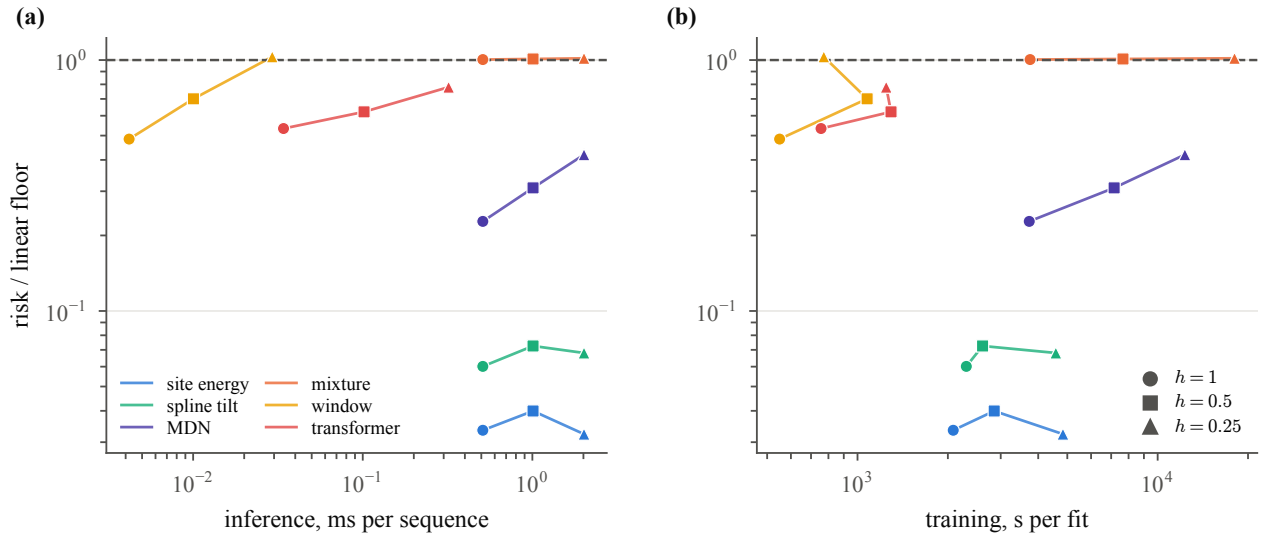


Figure 4: Error over the linear floor against (a) inference time per sequence and (b) training time per fit, at $\Theta = 0.3$ on $h = 1, \frac{1}{2}, \frac{1}{4}$ (circle, square, triangle; 32, 64, 128 sites), $M = 2048$. Inference: eight threads, batch 512, the largest network variants. Training: shared nodes, indicative only. BP's time is the same for every kernel.

- **Physics** (Figure 3): on the base cell $\hat{\Theta}/\Theta = 1.000 \pm 0.023$ and barrier 0.247 ± 0.011 against 0.25 (mean $\pm$ s.d. over sixteen seeds, $M = 2048$). At $\Theta = 2$ the barrier is small against the thermal energy and is learned slowly: its error falls like $M^{-0.54}$ and all sixteen seeds are within 20% at $M = 32\,768$, so it is a sample-size limit, not a floor.

4 Your predictions: refinement, cost, speciation

Refinement. The correlation length in sites grows like $1/h$ (6.1 to 22.5 from $h = 1$ to $\frac{1}{4}$). The window CNN's radius was screened on each lattice and the screen picked 2, 4, 8 sites, a constant physical reach (8 was the largest offered). Trained on noisy copies (ISM) with $M = 2048$, its absolute error falls ($0.149 \to 0.118$), but relative to the spline it grows from 9.2 to 17.6, and relative to the floor from 0.55 to 1.20: on the finest lattice the window is worse than the Gaussian closure. Per step the field becomes nearly Gaussian as $h \to 0$ (your $O(h)$ caveat), the floor falls ($0.269 \to 0.098$), and the site energy still stays at 3–4% of it: the structure builds up over many sites, and BP collects it. Whether a wider window closes the gap was not tested.

Cost. Figure 4: the networks own the cheap end (0.004–0.32 ms per sequence) and the structured kernels the accurate end (0.51–2.0 ms for BP), with nothing in between. The grid is the lever. With the true ϕ^4 kernel, BP on 101 points is exact to 1.8×10^{-11}; the learned site energy and spline refitted on 101 points have the same error as on 401 to three digits (at most 1.2% off at 51), the EM fit is about 14 times faster, and on one thread BP takes 0.107 ms per sequence at 101 points against 0.667 ms at 401. The paper's mixtures need about 151 points, because with much data they learn components narrower than a 101-point spacing.

Speciation. For a noisy snapshot the exact posterior gives each site $p_i = P(a_i > 0 \mid \mathbf{x})$; the commitment $|2p_i - 1|$ goes from 0 (both wells equally likely) to 1 (the site has chosen), and t_c is the level where its site average crosses $\frac{1}{2}$. All five registered predictions on commitment held (Figure 5):

- the posterior is calibrated: the share of signs guessed right equals $(1 + \text{commitment})/2$ to 0.003;
- colder fields commit earlier in the reverse process: t_c falls from 1.61 at $\Theta = 0.2$ to 0.46 at $\Theta = 2$;
- the exact posterior commits before its Gaussian twin at every temperature, by 0.47 in t when coldest and 0.07 when hottest. At $\Theta = 0.2$ it is half-committed when the noise variance is 24 times the signal's, the twin only at 8.9: the two humps, not just the correlations, decide when it commits;
- t_c grows with the chain length (1.03 at $L = 4$, 1.72 at 64) and stops once the chain is several domains long (1.750, 1.754 at 128, 256);

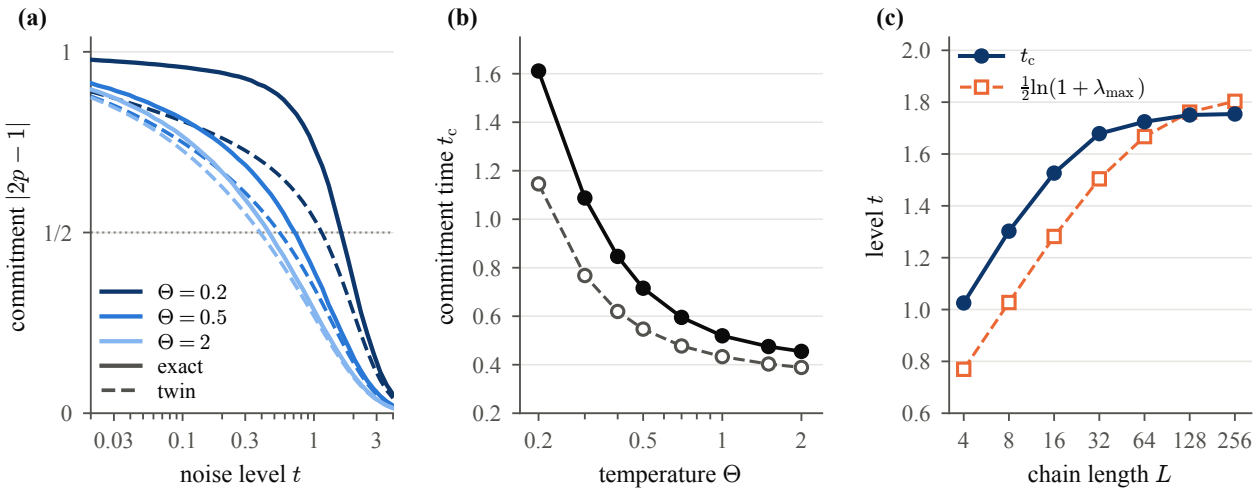


Figure 5: Commitment of the exact denoiser. (a) Site-averaged $|2p-1|$, with $p = P(a_i > 0 \mid \mathbf{x})$, against t: exact posterior (solid) and its Gaussian twin with the same covariance (dashed). (b) The level t_c where it crosses $\frac{1}{2}$, against Θ ($h = 1$, $L = 32$). (c) At $\Theta = 0.2$, t_c of the central sites against L, beside $\frac{1}{2}\ln(1+\lambda_{\max})$.

- below the correlation length, t_c moves one-for-one with $\frac{1}{2}\ln(1+\lambda_{\max})$ (slope 0.98), $\lambda_{\max}$ the largest covariance eigenvalue: the finite-chain analogue of the speciation time $\frac{1}{2}\ln\lambda_{\max}$ of Biroli, Bonnaire, de Bortoli and Mézard.

The dilute-kink law $\xi \propto \Theta^{1/2} e^{E_{\text{kink}}/\Theta}$*, fitted on* $\Theta \in [0.1, 0.3]$*, returns a barrier of* 0.988 *against* $2\sqrt{2}/3$ *(predicted within* 2%*).* The local slope approaches the theory as Θ falls, so the law looks right but these temperatures do not pin its constant; the lattice itself is converged (ξ moves by 0.04% from $h = 1/16$ to $1/32$).

5 Where it stops working, and where a neural kernel pays

Table 4: Stress tests. "Floor" is the best linear estimator. Sixteen seeds unless stated.

Test	Outcome
Observe every 4th site (clean fits; the kernel is K^4, no longer a tilted walk)	Pair energy and spline worsen 25× and 14× but stay far below the floor (0.040, 0.036 against 0.235). The MDN is below the site energy in 16 of 16 seeds, at 0.40 times its error; at stride 1 the site energy is 3.0 times better than the MDN.
A chain outside the class: a Laplace bond $\lvert a' - a\rvert$ on a double well (noisy fits, 8 seeds)	At $M = 2048$ the site energy stalls at 0.103; a 100-number unrestricted kernel reaches 0.068, the MDN 0.056 (below the site energy in 7 of 8 seeds).
The 100-number unrestricted kernel on the true field	Costs 2.5–5.5 times the pair energy. *The price grows with M; we predicted it would shrink.* Useful as a misspecification test, not as the estimator.
A hidden sign shared by the whole sequence (Gaussian test chain; the data are not Markov)	A first-order kernel stalls (+0.2% from 4 times more data) at 1.56–1.71 times the floor. BP on the augmented state (value, sign) repairs it: 6.3 times below the floor.
Snapshots after a quench (not Gibbs; reference is the clean snapshot)	The site energy beats an empirical Gaussian smoother at every time (ratio 0.986 early, 0.950 at equilibrium); the spline adds nothing.
Missing data: hide sites, sample posteriors by FFBS	The learned site energy matches the oracle within 0.4% in error and interval width, coverage 0.890–0.893 for a nominal 0.90. On the cold cell a Gaussian smoother is only 1.5%–4.6% worse, with 7%–9% wider intervals.

So a neural kernel earns its parameters when the simple family is wrong (stride 4; the Laplace bond), by 1.8–2.5 times against the best Gaussian-bond kernel, and not when it is right. On the paper's chain it loses to the 25-number mixture in 64 of 64 comparisons. In both winning cases the MDN was still improving when

its budget ended (11 of 16 and 5 of 8 fits), so its numbers there are upper bounds.

6 What we do not claim

- **Speed.** BP is slower per sequence than a network. On the 401-point grid with eight threads it is 12.5× the transformer on CPU and 15.1× on an H200. On one thread the gap narrows (3.2× at 401 points) and at 101 points BP is faster than the transformer, but still slower than the GRU. The networks timed are the largest screened variants.
- **Stopping rule.** Every arm's stopping point or checkpoint, networks included, is picked with the exact score on a validation set. With an oracle-free rule for EM (held-out likelihood) the site energy is at 4–23% of the floor instead of 3–17%.
- **Small data.** With $M = 32$ on the two hottest cells the site energy is above the floor (1.37, 1.43); it wins on average from $M = 128$, and in 16 of 16 seeds from $M = 512$.
- **Failed predictions,** beyond those above: *at* $M = 128$, $\Theta = 2$ *the mixture beats the spline in 12 of 16 seeds; the 7-number Legendre tilt crashes on the finest lattice (39 failed fits); the barrier at* $\Theta = 1$ *is within* 20% *in 14 of 16 seeds, not 16; the information-decay theorem's* e^{-8t} *regime is not reached by* $t = 3$ *(ratio* 0.73–0.94*, unresolved); the raw count of sign changes is not a lattice-independent kink density (we count kinks after merging short runs).*
- **Unexplained:** why the unrestricted kernel's price over the site energy grows with data.

7 The 25 September questions

Answered in the order of the call. The linear chain is $a_{i+1} = \rho a_i + \varepsilon_i$, $\rho = 0.85$, Laplace driving noise, $L = 32$, unless stated.

Q1. Gaussian messages = Gaussian score = Kalman? Yes. On a linear chain, BP whose messages are projected onto Gaussians returns exactly the best linear estimator of the covariance-matched Gaussian chain, $\widehat{\mathbf{a}}_{\mathrm{LMMSE}}(\mathbf{x}) = e^{-t}\boldsymbol{\Sigma}(e^{-2t}\boldsymbol{\Sigma} + \Delta_t \mathbf{I})^{-1}\mathbf{x}$, which is the Rauch–Tung–Striebel smoother. The hypothesis that matters is that the first message is Gaussian (or projected) too; without it the claim is false, by 4×10^{-2} at the first site in a computed counterexample. Check: grid BP, the closed form and an independent RTS smoother agree to 1.9×10^{-13}. *Projected BP run on the same 401-point grid is off by up to* 0.26*: its backward messages widen past the grid's edge.* Not new: the thesis had it; the paper now says "Kalman".

Q2–Q4. Do messages stay in a finite family? Your family, a Gaussian pushed once through the chain, $\mathcal{N}(m, v)$ dilated by ρ and convolved with p_ε, is exact at the first site and very close afterwards: 2,272× smaller KL than a plain Gaussian and 78× smaller than a 4-parameter Gaussian+Laplace at low noise. But it is not closed, $(G * p_\varepsilon) \cdot \mathcal{N} \neq G' * p_\varepsilon$: the interior error is above the quadrature floor in 432 of 432 configurations, and does not move when the grid is doubled. The Gaussian closes; we have no proof about other families. *We predicted the family would do best at high noise; it does worst there,* because with a flat likelihood the recursion drives messages to the stationary law, an infinite convolution.

Q5. Fourier BP? Two of the three operations are free in Fourier space, which gives the message in closed form, $\varphi_{k+1}(\omega) = \varphi_\mu(\rho^k \omega) \prod_{j<k} \varphi_\varepsilon(\rho^j \omega)$ (checked against grid BP to 2×10^{-8}). The likelihood is the hard step, a convolution in ω of width $e^{-t}/\sqrt{\Delta_t}$, widest at low noise. As a representation, Fourier modes on a window adapted to each message need 33 numbers (KL 10^{-2}) or 47 (10^{-4}) against the grid's 45 and 151, close to your "20 modes"; but they reach 10^{-4} in only 53 of 144 cases and adding modes can make it worse, so a solver cannot choose a truncation. Cumulant (Gram–Charlier) truncation does not converge: Cramér's condition fails for every message, and adding cumulants makes the reconstruction worse in 144 of 144 cases. And the transition step is only 12.1%–15.2% of a sweep, so making it free buys at most 1.14–1.18×; our FFT version is 5.8× slower at 401 points. The coarse grid is the better trick (Section 4).

Q6. A neural network for the kernel? It trains through BP and loses where a small kernel is right: on the linear chain the 1,516-parameter MDN is worse than the 25-number mixture in 64 of 64 comparisons (2.1–2.9×) and worse than the best plain score network. *We predicted it would sit between the two.* It wins where the simple family is wrong (Section 5). This is the overfitting you predicted in the call.

Q7. Time per posterior mean? See Section 6: BP is slower (0.52 against 0.042 ms, eight threads; on the H200 15× at batch 512, 6× at 2048, because BP is latency-bound). What scales well is training: the EM M-step costs the same whatever M (the E-step is linear in M), because the kernel is evaluated on the grid only.

Q8. Generation without a score? Yes. On a chain, sampling site by site from the kernel is exact, and it is 731× cheaper than a 200-step reverse SDE at equal error (eight threads, 2,048 sequences). A Gaussian score generates Gaussian samples (excess kurtosis of $a_{i+1} - 0.85a_i$ about 0 against 1.92).

Q9. Fixed t or pooled? Per level: EM–BP beats every network at all twelve levels, and beats the Gaussian oracle at the 8 ($M = 128$) or 10 ($M = 2048$) lowest. Above that the non-Gaussian structure left is smaller than the kernel's estimation error; the crossover moves from $t \approx 1$ to $t \approx 2$ with 16 times more data. *We predicted first place at every level.*

Q10. Information about ρ, β or about κ_2, κ_4? Recomputed, not relabelled. With $\kappa_2 = q$, $\kappa_4 = \gamma_2(\beta)q^2$ and $J = \partial(\rho, \kappa_2, \kappa_4)/\partial(\rho, q, \beta)$, the efficiency of coordinate k is $1/(j_k^\top I_\theta^{-1} j_k)$; a relabel would be wrong by 1.6–1526×. The message survives: fourth-order information falls about 700× more than second order from $t = 0.05$ to $t = 3$. Following your advice it is a supporting result, not the headline.

Q11. Scaling in M and L? The call's answer was wrong. A sweep is $O(L\, N_\mathrm{g}^2 B)$ operations for B sequences: measured slopes 1.03 in B, 1.03 in L, and 1.16 (not 2) in N_g, because element-wise work and memory traffic dominate the matrix product. Memory: about 266 MB for 512 sequences of 32 sites, 1.8–4.2 times the largest network.

Your other requests. Figures first, with a summary figure: done (the draft opens with one). Labels of the information plot: re-plotted in cumulant coordinates. Notation: one table; "driving noise" instead of "innovation", "linear chain" instead of AR(1). Clean-input network arm: dropped.

To discuss

1. **Is this enough for a paper,** as you suggested: the general Markov kernel, the linear chain as the first example and the ϕ^4 field as the second? Venue and timeline.
2. **What to cut.** The draft is far over an 8-page body. Which five or six figures carry it?
3. **The headline.** "Learn the local law, infer globally; the kernel family is the modelling choice", with the neural kernel as the test of when flexibility pays?
4. **Marc.** How to involve him from here.
5. **The defence.** Can I show the field results as further work on Friday?

Sources: registry files `experiments/registry/E*.json` and the frozen outputs they list (field: E32–E34, E36–E37, E41–E50; questions: E22–E31, E39–E40, E47); full detail in `reports/field-laboratory-2026-10-01.pdf` and `docs/JEROME_CALL_2026-09-25.md`.